

MODULE 5

MORPHOLOGICAL OPERATIONS & REPRESENTATION AND DESCRIPTION

STRUCTURING ELEMENT - DILATION & EROSION

Set reflection and translation are employed extensively in morphology to formulate operations based on so-called structuring elements (SEs). Small sets or subimages used to probe an image under study for properties of interest.

The reflection of a set B , denoted $\hat{B}$, is defined as

$$\hat{B} = \{w/w = -b, \text{ for } b \in B\}$$

If B is the set of pixels (2D-points) representing an object in an image, then $\hat{B}$ is simply the set of points in B whose (x, y) coordinates have been replaced by $(-x, -y)$.

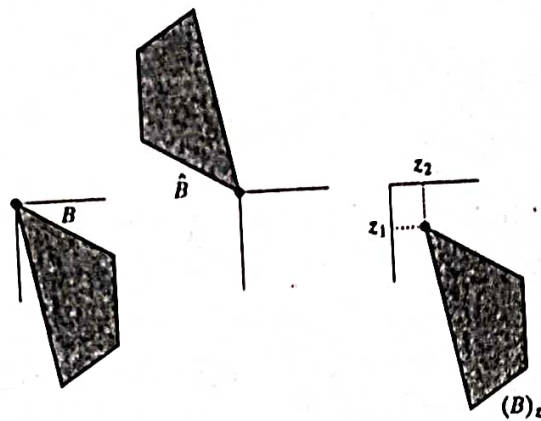
The translation of a set B by point $z = (z_1, z_2)$ denote $(B)_z$ is defined as

$$(B)_z = \{c \mid c = b + z, \text{ for } b \in B\}$$

If B is the set of pixels representing an object in an image, then $(B)_z$ is the set of points in B whose (x, y) coordinates have been replaced by $(x+z_1, y+z_2)$

FIGURE 9.1

(a) A set, (b) its reflection, and (c) its translation by z .



The first row of the following figure shows several examples of structural elements where each shaded square denotes a member of the SE.

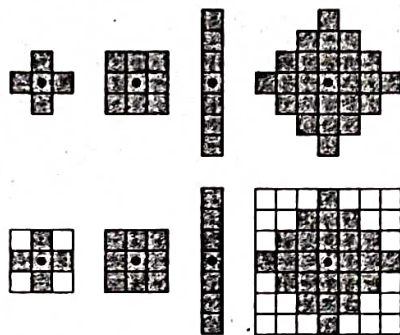


FIGURE 9.2 First row: Examples of structuring elements. Second row: Structuring elements converted to rectangular arrays. The dots denote the centers of the SEs.

Morphological Image Processing

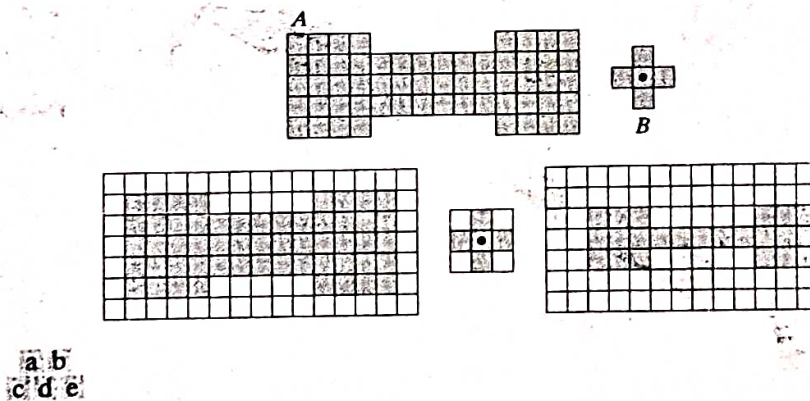


FIGURE 9.3 (a) A set (each shaded square is a member of the set). (b) A structuring element. (c) The set padded with background elements to form a rectangular array. (d) Structuring element as a rectangular array. (e) The result of processing the set from (c) with the structuring element from (d).

Erosion and dilation are two fundamental operations in morphological processing.

1. Erosion

With A and B as sets in $\mathbb{Z}^2$, the erosion of A by B , denoted $A \ominus B$, is defined as

$$A \ominus B = \{z \mid (B)_z \subseteq A\}$$

In words, this equation indicates that the erosion of A by B is the set of all points z such that B , translate by z , is contained in A .

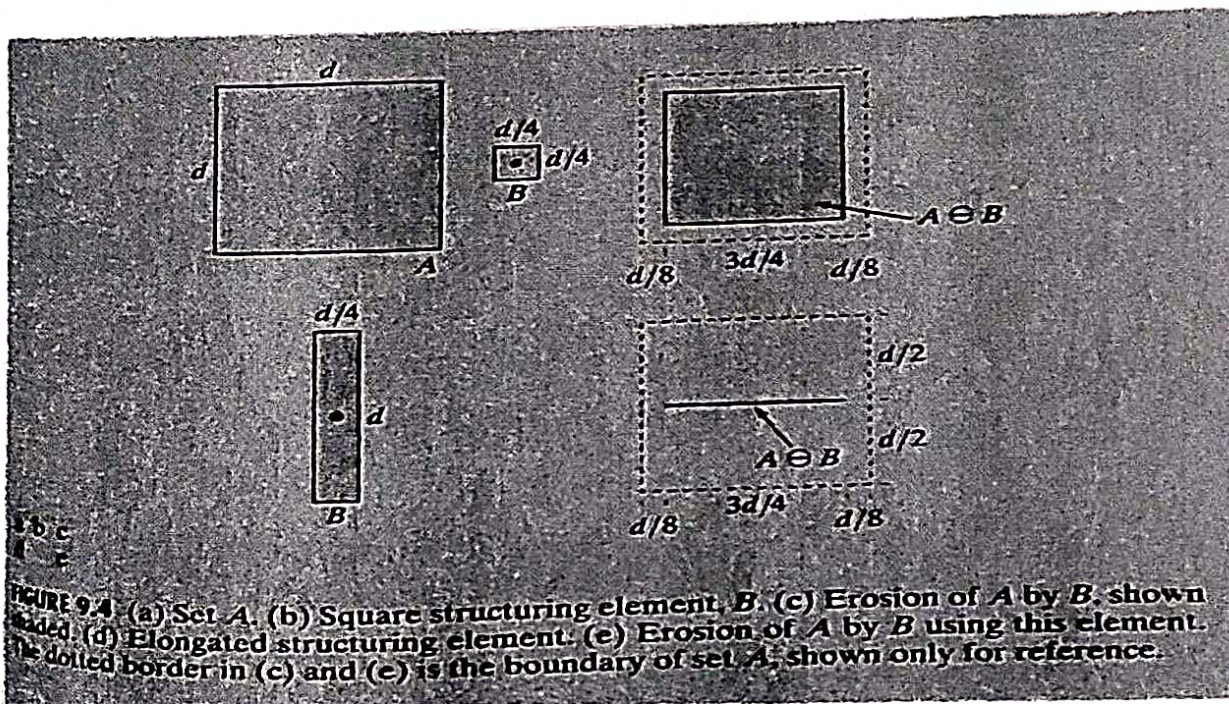
Equivalent form:

$$A \ominus B = \{z \mid (B)_z \cap A^c = \emptyset\}$$

A^c is the complement of A and

$\emptyset$ is the empty set.

Example of Erosion is shown in the following figure.



2. Dilation

With A and B as sets in $\mathbb{Z}^2$, the dilation of A by B, denoted $A \oplus B$ is defined as

$$A \oplus B = \{z \mid (\hat{B})_z \cap A \neq \emptyset\}$$

This equation is based on reflecting B about its origin, and shifting this reflection by z.

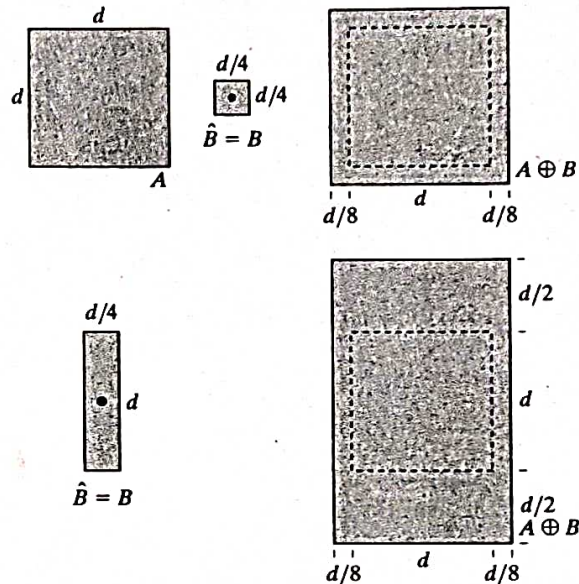
Equivalent expression for Dilation:

$$A \oplus B = \{z \mid [(\hat{B})_z \cap A] \subseteq A\}$$

a b c

FIGURE 9.6

- (a) Set A .
 (b) Square structuring element (the dot denotes the origin).
 (c) Dilation of A by B , shown shaded.
 (d) Elongated structuring element.
 (e) Dilation of A using this element. The dotted border in (c) and (e) is the boundary of set A , shown only for reference



An illustration of dilation

a b

FIGURE 9.7

- (a) Sample text of poor resolution with broken characters (see magnified view).
 (b) Structuring element.
 (c) Dilation of (a) by (b). Broken segments were joined.

Historically, certain computer programs were written using only two digits rather than four to define the applicable year. Accordingly, the company's software may recognize a date using "00" as 1900 rather than the year 2000.

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0	1	0
1	1	1
0	1	0

MORPHOLOGICAL OPENING AND CLOSING

Two other important morphological operations are : opening and closing.

Opening:

It generally smoothes the contour of an object, break narrow isthmuses, and eliminates thin protrusions.

Closing:

Closing also tends to smooth section of contours, but as opposed to opening, it generally fuses narrow breaks and long thin gulfs, eliminates small holes and fills gaps in the contour.

The opening of set A by structuring element B denoted $A \circ B$, is defined as

$$A \circ B = (A \ominus B) \oplus B$$

Thus, the opening A by B is the erosion of A by B followed by a dilation of the result by B .

Similarly, the closing of set A by structuring element B , denoted $A \bullet B$, is defined as

$$A \bullet B = (A \oplus B) \ominus B$$

The closing of set A by B is simply the dilation of A by B , followed by the erosion of the result by B .

The opening can be expressed as a fitting process such that

$$A \circ B = \bigcup \{ (B)_z \mid (B)_z \subseteq A \}$$

Where $\bigcup \{ \cdot \}$ denotes the union of all the sets inside the braces.

Geometrically, a point w is an element of $A \circ B$ if and only if $(B)_z \cap A \neq \emptyset$ for any translate of $(B)_z$ that contains w .

A simple geometric interpretation is shown in the following figure.

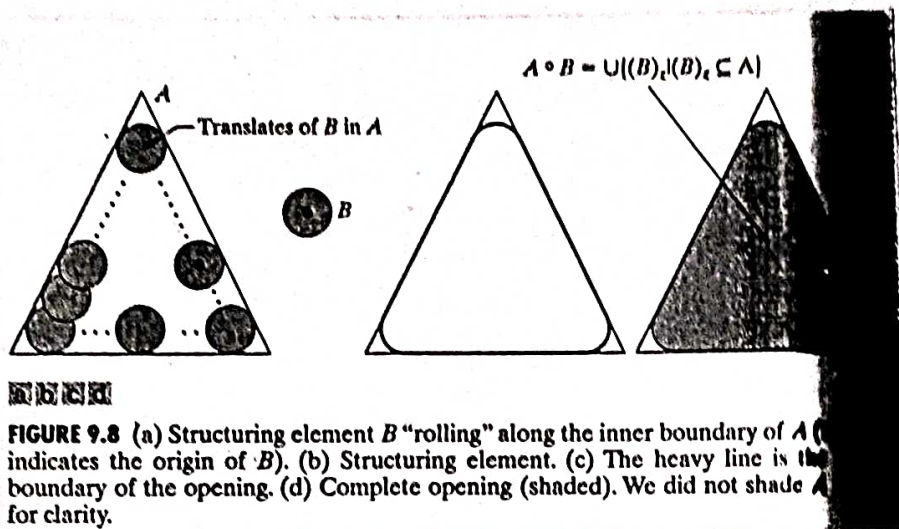
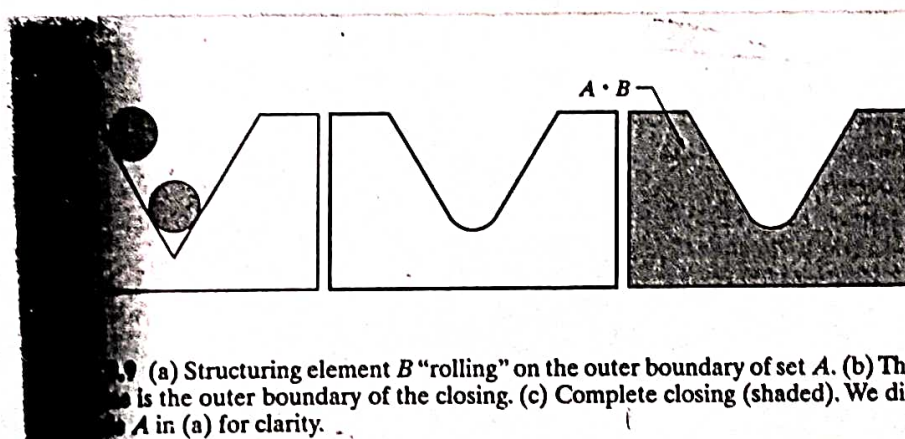


FIGURE 9.8

(a) Structuring element B "rolling" along the inner boundary of A indicates the origin of B . (b) Structuring element. (c) The heavy line is the boundary of the opening. (d) Complete opening (shaded). We did not shade A for clarity.

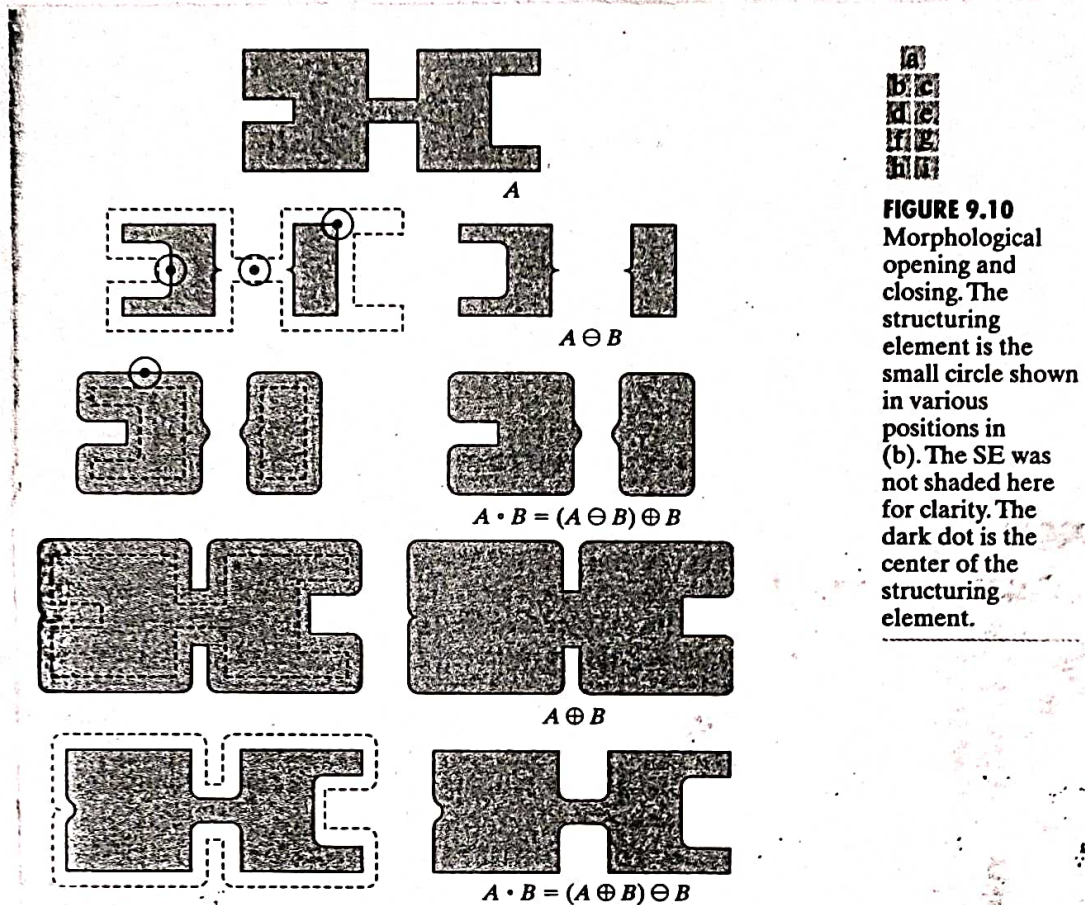
The following figure illustrates the basic geometrical properties of closing.

Fig. 9.9 G637



(a) Structuring element B "rolling" on the outer boundary of set A . (b) The heavy line is the outer boundary of the closing. (c) Complete closing (shaded). We did not shade A in (a) for clarity.

The following figure further illustrates the opening and closing operations.



The opening operation satisfies the following properties

(a) $A \circ B$ is a subset (subimage) of A .

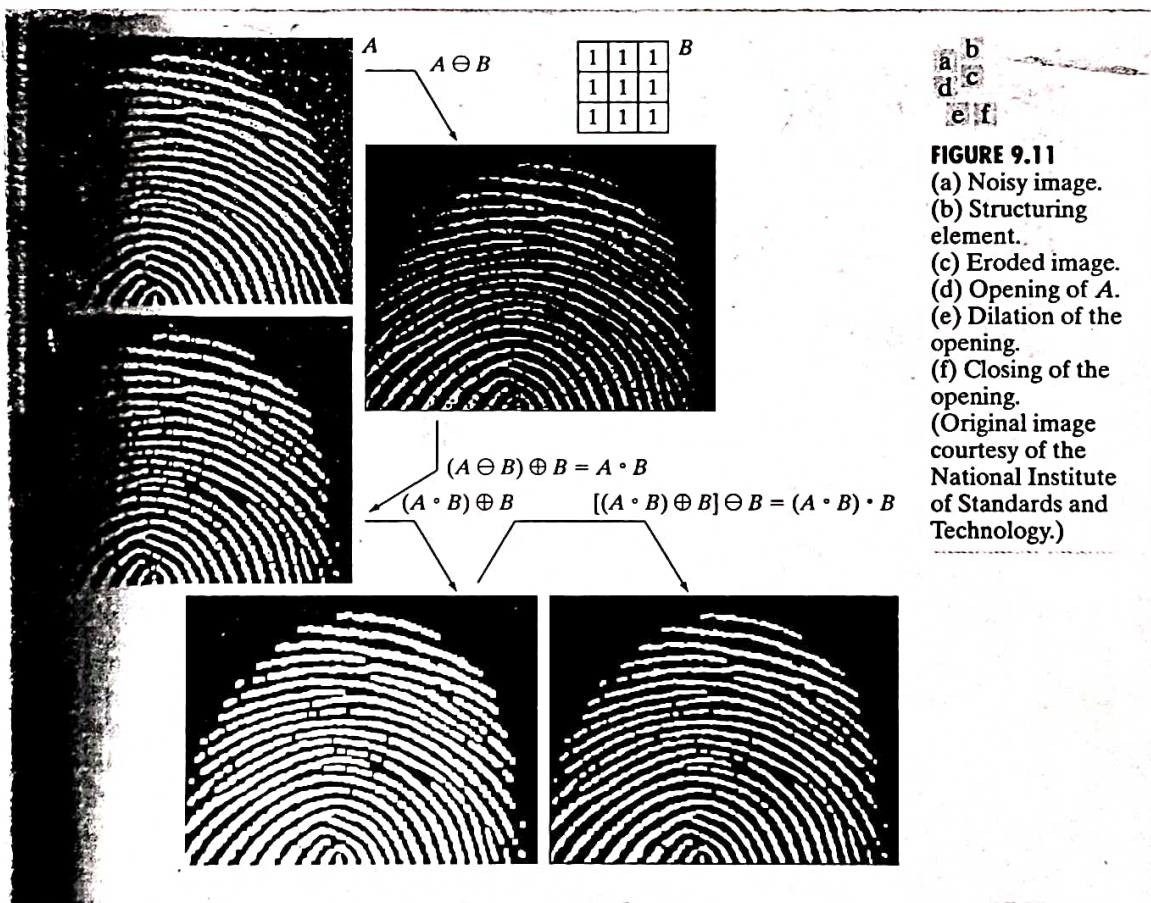
(b) If C is a subset of D , then $C \circ B$ is a subset of $D \circ B$.

(c) $(A \circ B) \circ B = A \circ B$



Similarly, the closing operation satisfies the following properties:

- (a) A is a subset (subimage) of $A \cdot B$.
- (b) If C is a subset of D , then $C \cdot B$ is a subset of $D \cdot B$.
- (c) $(A \cdot B) \cdot B = A \cdot B$



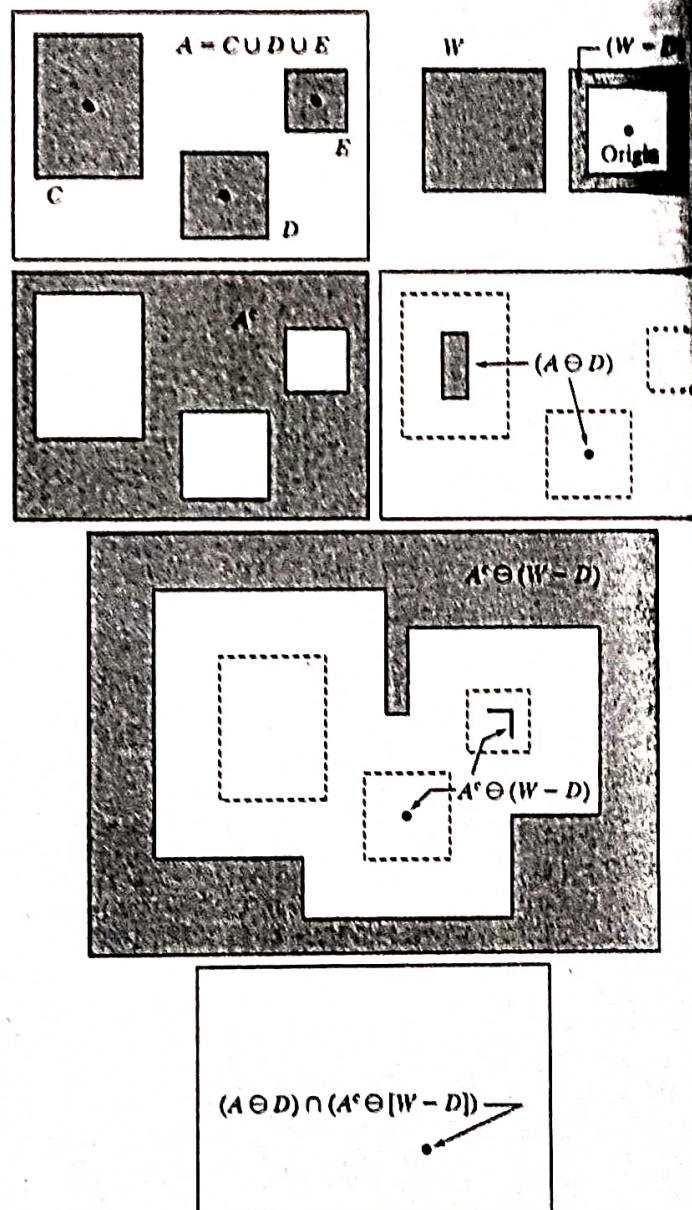
HIT OR MISS TRANSFORMATION

The morphological hit-or-miss transform is a basic tool for shape detection.

a b
c d
e f

FIGURE 9.12

(a) Set A . (b) A window W , and the local background of D with respect to W , $(W - D)$. (c) Complement of A . (d) Erosion of A by D . (e) Erosion of A^c by $(W - D)$. (f) Intersection of (d) and (e), showing the location of the origin of D , as desired. The dots indicate the origins of C , D , and E .





B denotes the set composed of D and its background, the match (or set of matches) of B in A , denoted $A \otimes B$ is,

$$A \otimes B = (A \ominus D) \cap [A^c \ominus (W-D)] \quad (1)$$

We can generalize the notation somewhat by letting $B = (B_1, B_2)$, where B_1 is the set formed from elements of B associated with an object and B_2 is the set of elements of B associated with the corresponding background.

From the preceding discussion, $B_1 = D$ and $B_2 = (W-D)$.

With this notation Eq(1) becomes

$$A \otimes B = (A \ominus B_1) \cap (A^c \ominus B_2) \quad (2)$$

Thus, set $A \otimes B$ contains all the (origin) points at which, simultaneously, B_1 found a match (hit) in A and B_2 found a match in A^c .



By using the definition of set differences and the dual relationship between erosion and dilation, we can write Eq(2) as in Eq (3).

$$A - B = \{w | w \in A, w \notin B\} = A \cap B^c$$

$$(A \ominus B)^c = A^c \oplus \hat{B}$$

$$A \circledast B = (A \ominus B_1) - (A \oplus \hat{B}_2)$$

————— (3)

However, Eq(2) is considerably more intuitive. We refer to any of the preceding three equations as the morphological hit-or-miss transforms.



BOUNDARY (BORDER) FOLLOWING

A boundary following algorithm gives an ordered sequence of points as output.

Given a binary region R on its boundary, an algorithm for following the border of R , on the given boundary consists of the following steps:

1. Let the starting point, b_0 , be the uppermost, leftmost point in the image that is labeled 1. Denote by c_0 the west neighbor of b_0 . Clearly, c_0 always is a background point. Examine the 8-neighbors of b_0 , starting at c_0 and proceeding in a clock wise direction. Let b_1 denote the first neighbor encountered whose value is 1, and let c_1 be the point immediately preceding b_1 in the the sequence. Store the locations of b_0 and b_1 for use in step 5. [Refer the following figure]

2. Let $b = b_1$ and $c = c_1$ [fig (c)]

3. Let the 8-neighbors of b , starting at c and proceeding in a clockwise direction, be denoted by



$n_1, n_2, \dots, n_k$. Find the first n_k labeled 1.

4. Let $b = n_k$, and $c = n_{k-1}$.

5. Repeat step 3 and 4 until $b = b_0$ and the next boundary point found is b_1 . The sequence of b points found when the algorithm stops constitutes the set of ordered boundary points.

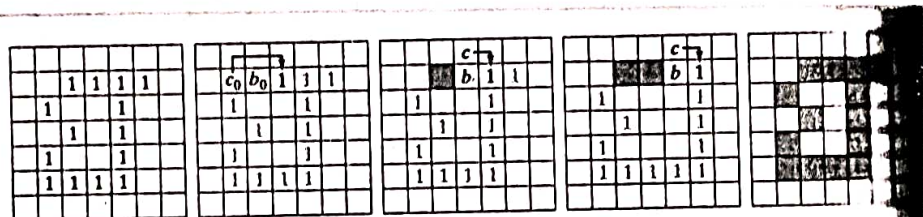


FIGURE 11.1

Illustration of the first few steps in the boundary-following algorithm. The point to be processed next is labeled in black, the points yet to be processed are gray, and the points found by the algorithm are labeled as gray squares.

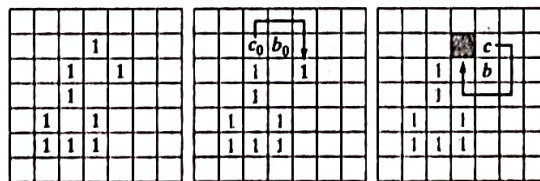


FIGURE 11.2

Illustration of an erroneous result when the stopping rule is such that boundary-following stops when the starting point, b_0 , is encountered again.

fig 11.2



CHAIN CODES

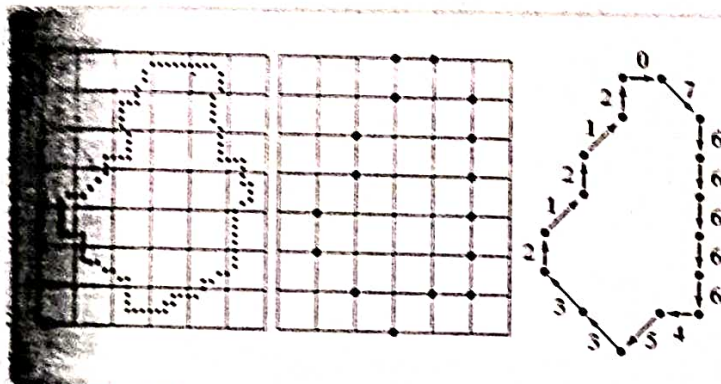
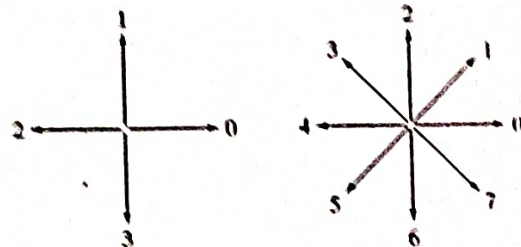
Chain codes are used to represent a boundary by a connected sequence of straight-line segments of specified length and direction. Typically this representation is based on 4- or 8-connectivity of the segments. The direction of each segment is coded by using a number scheme as shown in the following figure. A boundary code formed as a sequence of such directional numbers is referred to as a Freeman chain code.

This method is unacceptable due to the following reasons:

1. The resulting chain tends to be quite long
2. Any small disturbances along the boundary due to noise or imperfect segmentation cause changes in the code that may not be related to the principal shape features of the boundary.

FIG 11.3

FIGURE 11.3
Direction numbers for
(a) 4-directional
chain code, and
(b) 8-directional
chain code.



a b c

FIGURE 11.4
(a) Digital
boundary with
resampling grid
superimposed.
(b) Result of
resampling.
(c) 8-directional
chain-coded
boundary.



POLYGONAL APPROXIMATIONS

POLYGONAL APPROXIMATIONS USING MINIMUM PERIMETER POLYGONS.

A digital boundary can be approximated with arbitrary accuracy by a polygon. The goal of a polygonal approximation is to capture the essence of the shape in a given boundary using the fewest possible number of segments. Among several methods one of the most powerful is representing a boundary by a minimum-perimeter polygon (MPP).

1. Foundation:

An intuitively appealing approach for generating an algorithm to compute MPPs is to enclose a boundary shown in below figure (a) by a set of concatenated cells, as shown in figure (b).

The size of the cells determines the accuracy of the polygonal approximation. The error in each cell between



the boundary and the MPP approximation at most would be $\sqrt{2}d$, where d is the minimum possible distance between pixels.

MPP Algorithm

* The set of cells enclosing a digital boundary is called a cellular complex.

* Assume that the boundaries under consideration are not self intersecting, which leads to simply connected cellular complexes.

* Based on these assumptions, and letting white (W) and black (B) denote convex and mirrored concave vertices, respectively, they made some observations:

1. The MPP bounded by a simply connected cellular complex is not self intersecting.
2. Every convex vertex of the MPP is a W vertex, but not every B of a boundary is a vertex of the MPP.
3. Every mirrored concave vertex of the MPP is a B vertex, but not every B vertex of a boundary is a vertex of the MPP.



4. All B vertices are on or outside the MPP, and all W vertices are on or inside the MPP.
5. The uppermost, leftmost vertex in a sequence of vertices contained in a cellular complex is always W vertex of the MPP.

BOUNDARY DESCRIPTORS

1. Some Simple Descriptors.

The length of a boundary is one of its simplest descriptors. The number of pixels along a boundary gives a rough approximation of its length.

The diameter of a boundary B is defined as

$$\text{Diam}(B) = \max_{i,j} [D(P_i, P_j)]$$

Where D : distance measure

P_i, P_j : Points on the boundary.

2. Shape Numbers

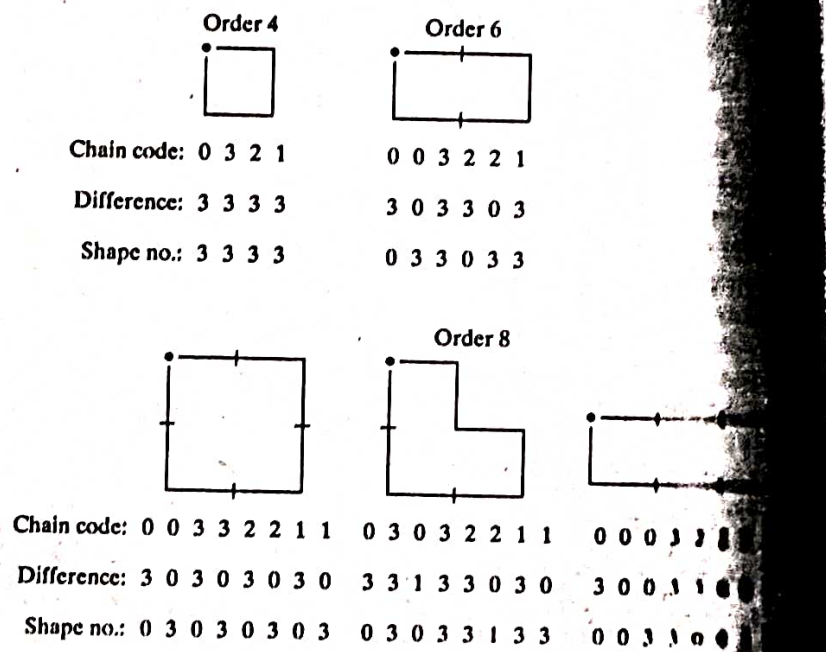
The shape number of such a boundary, based on the 4-directional code, is defined as the first difference of smallest magnitude.

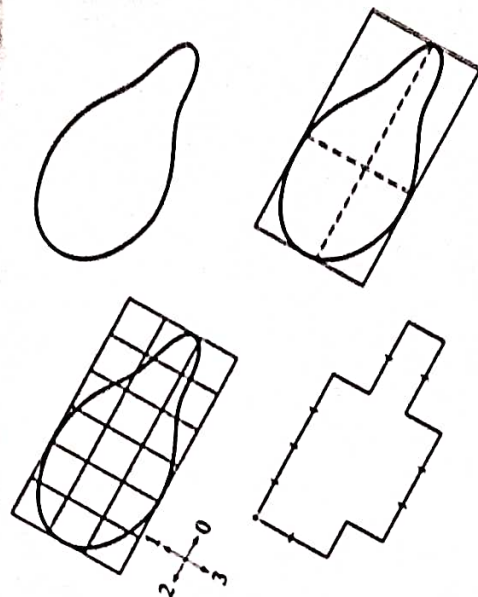
* The order 'n' of a shape number is defined as the number of digits in its representation.

* Moreover, n is even for a closed boundary, and its value limits the number of possible different shapes.

FIGURE 11.17

All shapes of order 4, 6, and 8. The directions are from Fig. 11.3(a), and the dot indicates the starting point.





a b
c d

FIGURE 11.18
Steps in the
generation of a
shape number.

Chain code: 0 0 0 0 3 0 0 3 2 2 3 2 2 2 1 2 1 1

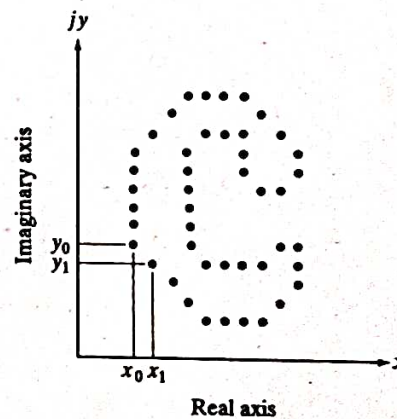
Difference: 3 0 0 0 3 1 0 3 3 0 1 3 0 0 3 1 3 0

Shape no.: 0 0 0 3 1 0 3 3 0 1 3 0 0 3 1 3 0 3

2. Fourier Descriptors.

FIGURE 11.19

A digital boundary and its representation as a complex sequence. The points (x_0, y_0) and (x_1, y_1) shown are (arbitrarily) the first two points in the sequence.





The above figure shows a k -point digital boundary in the xy -plane. Starting at an arbitrary point (x_0, y_0) , coordinate pairs (x_0, y_0) , (x_1, y_1) , (x_2, y_2) , $\dots$, (x_{k-1}, y_{k-1}) are encountered in traversing the boundary, in the counter clockwise direction.

$$x(k) = x_k \quad \text{and} \quad y(k) = y_k$$

$$s(k) = [x(k), y(k)], \quad \text{for } k = 0, 1, 2, \dots, k-1.$$

Each coordinate pair can be treated as a complex number so that

$$s(k) = x(k) + j y(k).$$

$$\text{for } k = 0, 1, 2, \dots, k-1.$$

→ x -axis is treated as real axis.

→ y -axis is treated as imaginary axis.

The DFT of $s(k)$ is

$$a(u) = \sum_{k=0}^{k-1} s(k) e^{-j 2\pi u k / k}$$

$$\text{for } u = 0, 1, 2, \dots, k-1.$$

The complex coefficients $a(u)$ are called the Fourier descriptors of the boundary.



The inverse Fourier transform of these coefficients restores $s(k)$

$$s(k) = \frac{1}{K} \sum_{u=0}^{K-1} a(u) e^{j2\pi uk/K}$$

where $a(u) = 0$ for $u > P-1$.

for $k = 0, 1, 2, \dots, K-1$.

Instead of all the Fourier coefficients, only the first P coefficients are used.

Approximation to $s(k)$, as $\hat{s}(k)$

$$\hat{s}(k) = \frac{1}{P} \sum_{u=0}^{P-1} a(u) e^{j2\pi uk/P}$$

for $k = 0, 1, 2, \dots, K-1$.

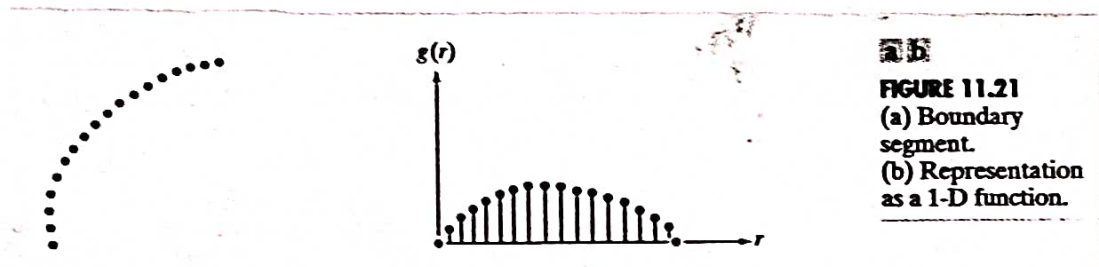
The following table summarizes the Fourier descriptors for a boundary sequence $s(k)$ that undergoes rotation, translation and changes in starting point.

TABLE 11.1
Some basic properties of Fourier descriptors.

Transformation	Boundary	Fourier Descriptors
Identity	$s(k)$	$a(u)$
Rotation	$s_r(k) = s(k)e^{j\theta}$	$a_r(u) = a(u)e^{j\theta}$
Translation	$s_t(k) = s(k) + \Delta_{xy}$	$a_t(u) = a(u) \cdot \Delta_{xy}$
Scaling	$s_s(k) = \alpha s(k)$	$a_s(u) = \alpha a(u)$
Starting point	$s_p(k) = s(k - k_0)$	$a_p(u) = a(u)e^{-j2\pi k_0 u/K}$

4. Statistical Moments.

The shape of boundary segments can be described quantitatively by using statistical moments such as the mean, variance and higher order moments. The accomplishment is shown in the following figure.



Let us treat the amplitude of g as a discrete random variable v and form an amplitude histogram $p(v_i)$, $i = 0, 1, 2, \dots, A-1$ where A is the number of discrete amplitudes increments in which we divide the amplitude scale. The n th moment of v about its mean is $\mu_n(v)$

$$\mu_n(v) = \sum_{i=0}^{A-1} (v_i - m)^n p(v_i)$$

Where $m = \sum_{i=0}^{A-1} v_i P(v_i)$

An alternative approach is to normalize $g(x)$ to unit area and treat it as a histogram. In this case, x is treated as the random variable and the moments are

$$\mu_n(x) = \sum_{i=0}^{K-1} (x_i - m)^n g(x_i)$$

where

$$m = \sum_{i=0}^{K-1} x_i g(x_i)$$

Where K : no. of points on the boundary

$\mu_n(x)$ directly relate to the shape of $g(x)$.

$g(x_i)$ probability of value x_i occurring

REGIONAL DESCRIPTORS

Various approaches for describing ~~image~~ regions are discussed in this. It is common practice to use both boundary and regional descriptors combined.



1. Some simple Descriptors

★ The area of a region is defined as the number of pixels in the region. The perimeter of a region is the length of its boundary.

★ A more frequent use of these two descriptors is in measuring compactness of a region, defined as $(\text{perimeter})^2 / \text{area}$.

★ A slightly different descriptor of compactness is the circularity ratio, defined as the ratio of the area of a region to the area of a circle having the same perimeter.

★ The area of a circle with perimeter length P is $P^2 / 4\pi$.

★ The circularity ratio, R_c is given by the expression

$$R_c = \frac{4\pi A}{P^2}$$

Where A is the area of the region

P is the length of its perimeter.

2. Topological Descriptors.

* Topological properties are useful for global descriptions of regions in the image plane.

- Simply defined, topology is the study of properties of a figure that are unaffected by any deformation, as long as there is no tearing or joining of the figure.

- Thus if a topological descriptor is defined by the number of holes in the region,

* Another topological property useful for region description is the number of connected components.

- A connected component of a region is defined as: For any pixel p in S , the set of pixels that are connected to it in S is called a connected component of S . Where S represents a subset of pixels in an image.

$$E = C - H$$

Where E : Euler number E

C : Connected components.

H : Number of holes.

The Euler number is also a topological property.

* Regions represented by a straight-line segments (referred to as polygonal networks) have a particularly simple interpretation in terms of the Euler number.

$$\text{Euler Formula} \Rightarrow V - Q + F = C - H$$

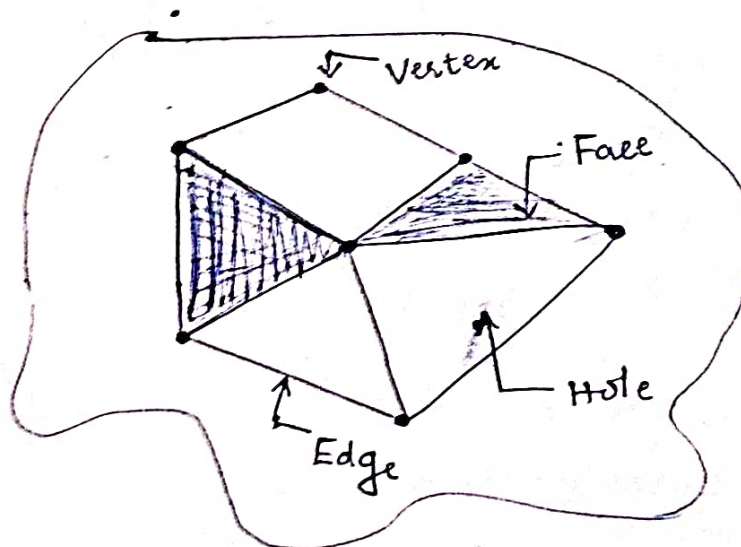
$$V - Q + F = C - H$$

$$V - Q + F = E$$

Where V : Number of vertices

Q : Number of edges

F : Number of faces.



In this figure we have



7 vertices (V)

11 Edges (Q)

2 Faces (F)

1 connected region (C)

3 holes (H)

$$\begin{aligned}\therefore \text{Euler number } E &= C - H \\ &= 1 - 3 \\ &= \underline{\underline{-2}}\end{aligned}$$

$$V - Q + F = C - H$$

$$7 - 11 + 2 = 1 - 3$$

$$\underline{\underline{-2}} = \underline{\underline{-2}}$$

3. Texture

An important approach to region description is to quantify its texture content. The three principal approaches used in image processing to describe the texture of a region are:

1. Statistical
2. Structural
3. Spectral



1. Statistical Approaches

Statistical approaches yield characterizations of textures as smooth, coarse, grainy and so on.

Let z be a random variable denoting intensity and let $p(z_i)$, $i = 0, 1, 2, \dots, L-1$, be the corresponding histogram, where L is the number of distinct intensity levels.

The n^{th} moment of z about the mean is

$$\mu_n(z) = \sum_{i=0}^{L-1} (z_i - m)^n p(z_i)$$

Where m is the mean value of z .

$$m = \sum_{i=0}^{L-1} z_i p(z_i)$$

$$\mu_0(z) = 1$$

$$\mu_1(z) = 0$$

$$\mu_3(z) = \sum_{i=0}^{L-1} (z_i - m)^3 p(z_i)$$

Measure of "uniformity" is given by

$$U(z) = \sum_{i=0}^{L-1} p^2(z_i)$$

Average entropy measure

$$e(z) = - \sum_{i=0}^{L-1} p(z_i) \log_2 p(z_i)$$



2. Structural Approaches.

Structural techniques deal with the arrangement of image primitives, such as the description of texture based on spaced parallel lines.

Suppose that we have a rule, of the form $S \rightarrow aS$ which indicates that the symbol S may be rewritten as aS .

The basic idea is that a simple "texture Primitive" can be used to form more complex texture Patterns by means of some rules that limit the number of possible arrangements of the primitives).

3. Spectral Approaches.

Spectral techniques are based on properties of the Fourier spectrum and are used primarily to detect global periodicity in an image by identifying high-energy, narrow peaks in the spectrum.

Three features of the Fourier Spectrum that are useful for texture description:

- (1) Prominent peaks in the spectrum give the principal direction of the texture patterns.



(2) The location of the peaks in the frequency plane gives the fundamental spatial period of the patterns.

(3) Eliminating any periodic components via filtering leaves non periodic image elements which can then be described by statistical techniques.

$$S(r) = \sum_{\theta=0}^{\pi} S_{\theta}(r) \quad \text{————— (1)}$$

Where S is the spectrum function

r and θ are variables in this coordinate s/m.

$S(r, \theta)$ may be considered a 1D function $S_{\theta}(r)$.

$$\text{and } S(\theta) = \sum_{r=1}^{R_0} S_r(\theta) \quad \text{————— (2)}$$

Where R_0 is the radius of a circle centered at the origin.

4. Moment Invariants

The 2-D moment of order $(p+q)$ of a digital image $f(x, y)$ of size $M \times N$ is defined as

$$m_{pq} = \sum_{x=0}^{M-1} \sum_{y=0}^{N-1} x^p y^q f(x, y)$$

Where $p = 0, 1, 2, \dots$ and $q = 0, 1, 2, \dots$ are integers.

The corresponding central moment of order $(p+q)$ is defined as

$$\mu_{pq} = \sum_{x=0}^{M-1} \sum_{y=0}^{N-1} (x - \bar{x})^p (y - \bar{y})^q f(x, y)$$

for $p = 0, 1, 2, \dots$ and $q = 0, 1, 2, \dots$ where

$$\bar{x} = \frac{m_{10}}{m_{00}} \quad \text{and} \quad \bar{y} = \frac{m_{01}}{m_{00}}$$

The normalized Central moments, denoted η_{pq} ,

are defined as $\eta_{pq} = \frac{\mu_{pq}}{\gamma}$

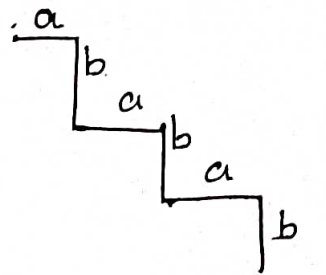
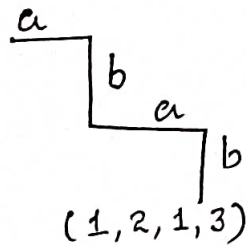
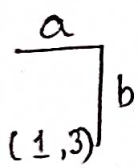
Where $\gamma = \frac{p+q}{2} + 1$

for $p+q = 2, 3, \dots$

(3) $A \rightarrow b$

Where S and A are variables and the elements a and b are constants corresponding to the primitives just defined.

Sample derivations for the rules $S \rightarrow aA$, $A \rightarrow bS$ and $A \rightarrow b$ are shown below.

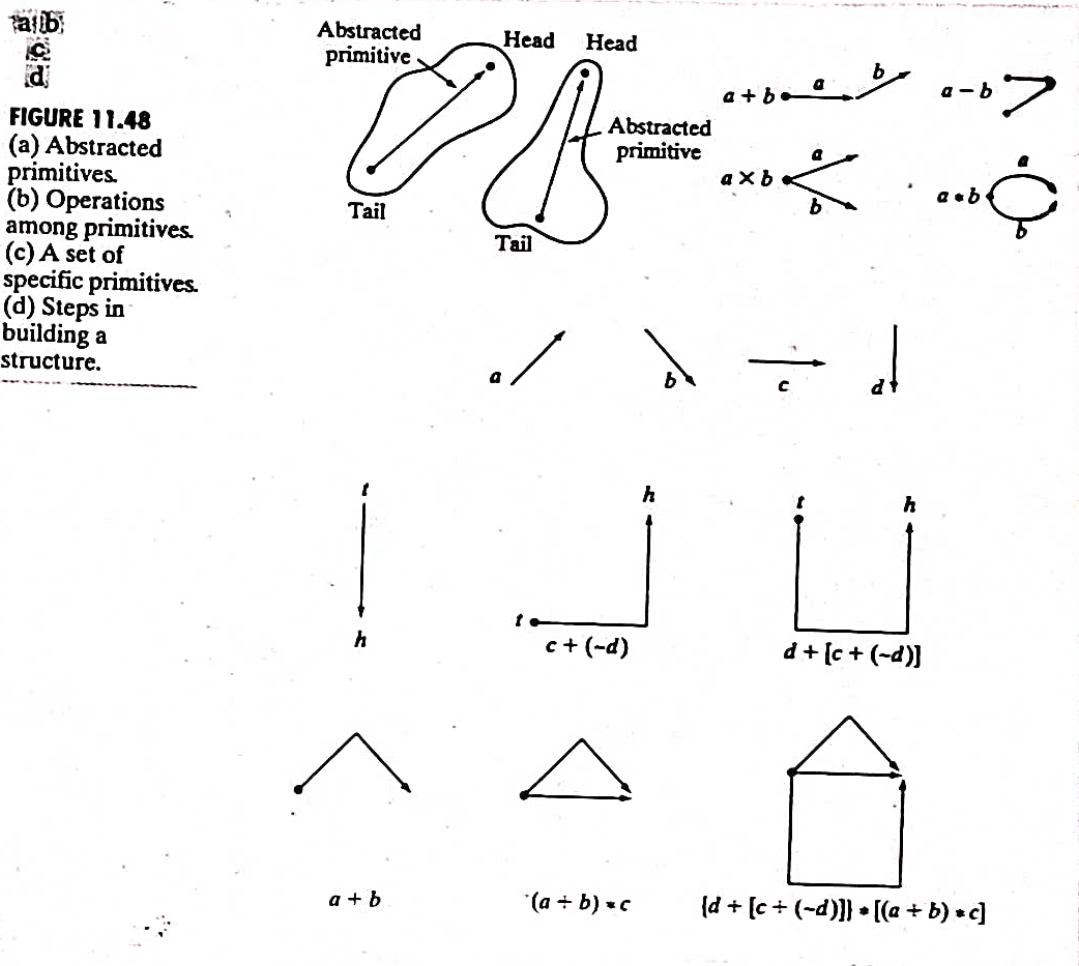


$(1, 2, 1, 2, 1, 3)$.

Another general approach is to describe sections of an image (such as small homogeneous regions) by directed line segments, which can be joined in other ways besides head-to-tail connections. This approach is shown in the following figure.

String descriptions are best suited for applications in which connectivity of primitives can be expressed in a head-to-tail or other continuous manner.

One of the most useful approaches for doing so is to use tree descriptors.



A tree T is a finite set of one or more nodes^{for} which

- (a) there is a unique node ϕ designated the root and
- (b) the remaining nodes are partitioned into 'm' disjoint sets $T_1, \dots, T_m$, each of which in turn is a tree called a subtree of T .



The tree frontier is the set of nodes at the bottom of the tree (the leaves), taken in order from left to right.

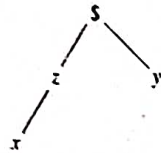
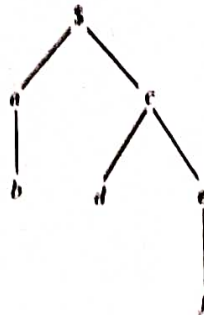
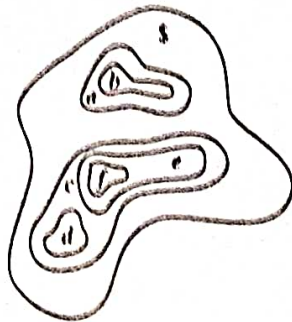


FIGURE 11.49 A simple tree with root s and frontier xy .

Generally, two types of information in a tree are important:

- (1) Information about a node stored as a set of words describing the node: identifies an image substructure (Eg:- region or boundary segment)
- (2) Information relating a node to its neighbors, stored as a set of pointers, to those neighbors: defines the physical relationship of that substructure to other substructures.



a/b

FIGURE 11.50
(a) A simple composite region.
(b) Tree representation obtained by using the relationship "inside of."